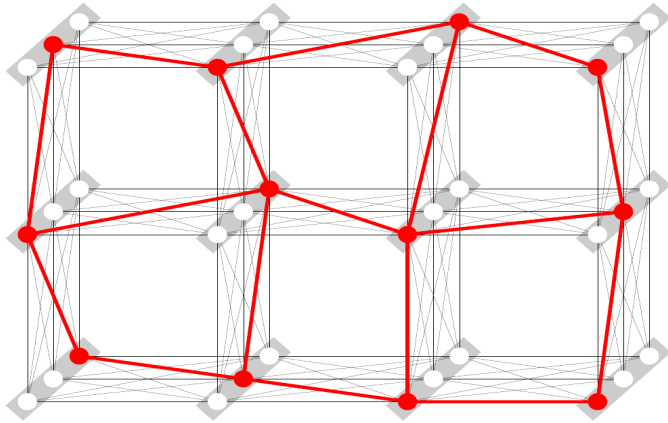


Combinatorial Solution Set

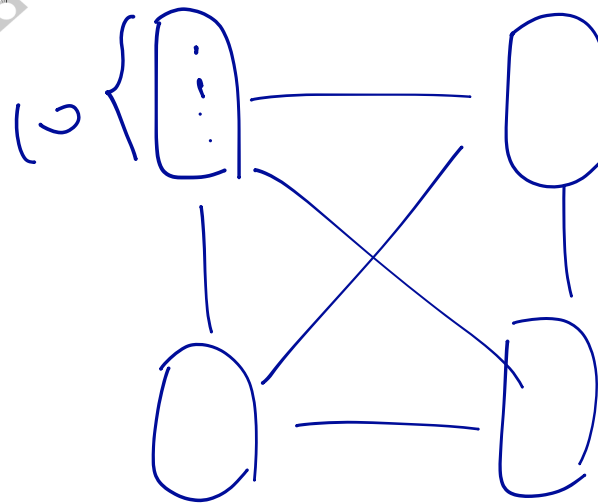
$$x^* = \min_{\bar{x} \in X_V} E_{\theta}(\bar{x}) := \min_{\bar{x} \in X_V} \sum_{v \in V} \theta_v(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}(x_u, x_v)$$



Number of possible labelings: $\prod_{v \in V} |X_V|$

2×2

Example: 4x4 grid and full graphs with 10 labels



- 1) 400
- 2) 440
- 3) 40
- 4) 1000
- 5) 10000
- 6) No correct answer

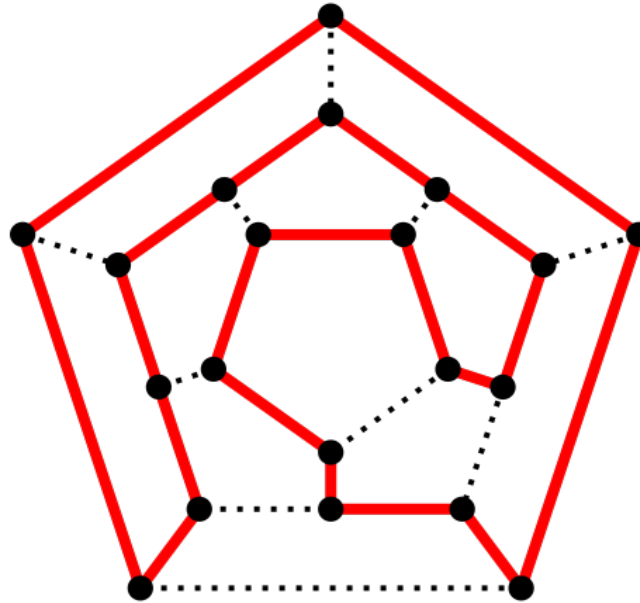
Picture: T Werner. A Linear Programming Approach to Max-sum Problem: A Review

Combinatorial Solution Set = No Polynomial Algorithm?

- 1) Finite state machine
- 2) Shortest path
- 3) 2-coloring for planar graphs
- 4) St-mincut
- 5) Linear assignment problem

Hamiltonian Cycle

A **Hamiltonian cycle** in a undirected graph ~~graph~~ is the cycle that visits each vertex exactly once.



Determining whether such a cycle exists in a graph is NP-complete.

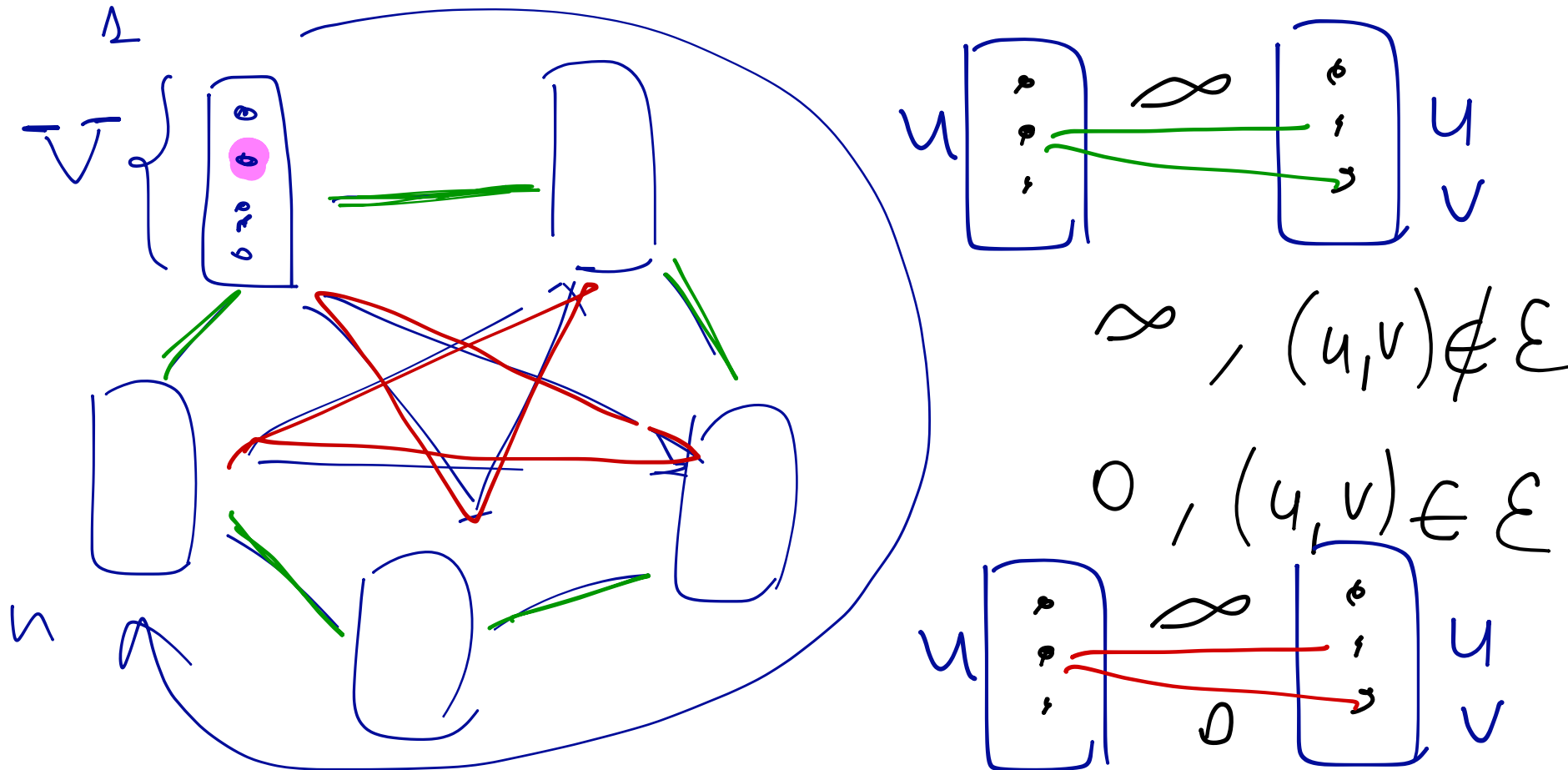
Picture: Wikipedia

Hamiltonian Cycle Reduces to MAP-Inference

Graph $(V, \mathcal{E})$ for a Hamiltonian cycle

Full graph $(V, V \times V)$ for the MAP-inference problem

The set of labels $X_v = V$ for all $v \in V$.



What about an inverse reduction:
MAP-inference to a Hamiltonian cycle?

How to Deal With NP-Hard Problems?

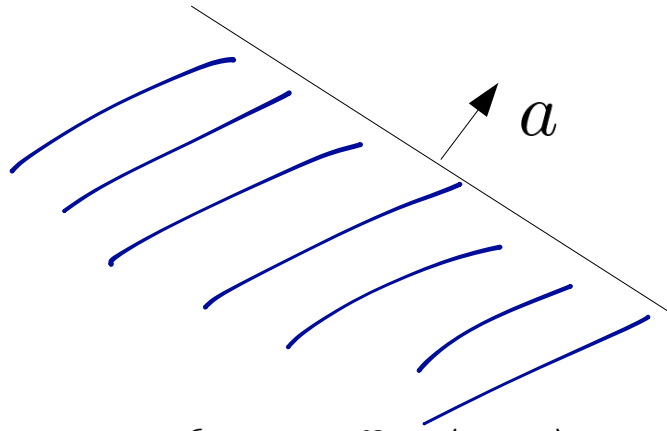
- 1) Identify polynomially solvable subclasses
- 2) Consider its convex relaxations (polynomially solvable approximations)
- 3) Develop approximate algorithms

We start with some background...

Linear Programs and Their Geometry

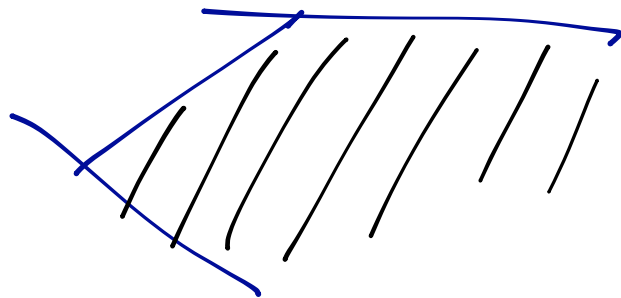
Integer linear programs is a standard way to describe combinatorial problems.

Hyperplane: $\{x \in \mathbb{R}^n : \langle a, x \rangle = b\}$

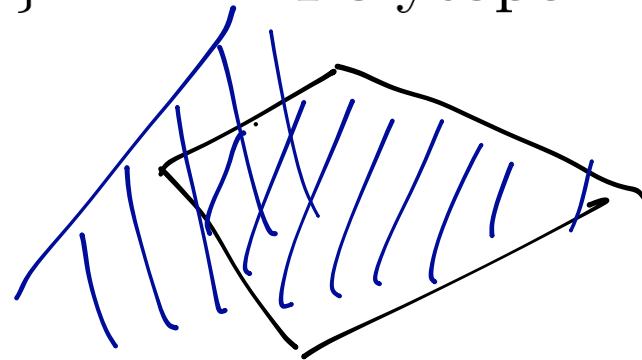


Half-space $\{x \in \mathbb{R}^n : \langle a, x \rangle \leq b\}$

Polyhedron: $P = \{x \in \mathbb{R}^n : Ax \leq b\}$



Polytope:



Linear Programs and Their Geometry

Polyhedron: $Ax \leq b$

Polyhedron? $Ax = b$
 $x_i \geq 0, i \in I \subset \{1, \dots, n\}$

1) Yes

2) No

3) Don't know

Linear Programs and Their Geometry

Polyhedron: $Ax \leq b$

Polyhedron? $Ax = b$ 1) Yes
 $x_i \geq 0, i \in I \subset \{1, \dots, n\}$ 2) No

$$\begin{aligned} Ax &\leq b \\ -Ax &\leq -b \\ -x_i &\leq 0, i \in I \end{aligned}$$

Linear Programs and Their Geometry

What about an inverse transformation:

$$Ax \leq b \quad \text{to} \quad \begin{array}{l} Ax = b \\ x_i \geq 0, \quad i \in I \subset \{1, \dots, n\} \end{array} \quad ?$$

Linear Programs and Their Geometry

What about an inverse transformation:

$$Ax \leq b \quad \text{to} \quad \begin{array}{l} Ax = b \\ x_i \geq 0, \quad i \in I \subset \{1, \dots, n\} \end{array} \quad ?$$

$$\begin{array}{l} Ax + x' = b \\ x' \geq 0 \end{array}$$

Linear Programs and Their Geometry

Linear programs (also holds for **max**):

Standard form

$$\min_{x \in \mathbb{R}^n} \langle c, x \rangle$$

$$Ax \leq b$$

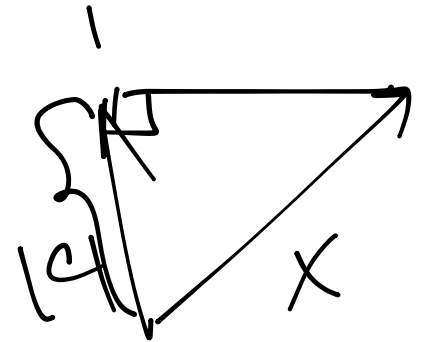
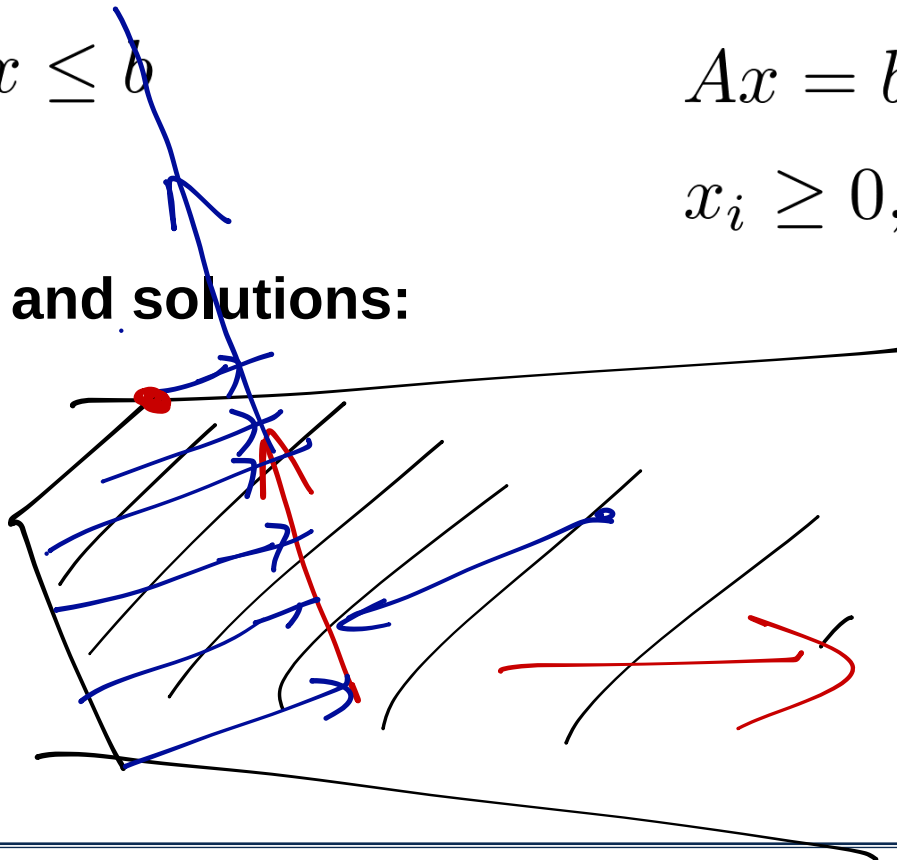
Canonical form

$$\min_{x \in \mathbb{R}^n} \langle c, x \rangle$$

$$Ax = b$$

$$x_i \geq 0, \quad i \in I \subset \{1, \dots, n\}$$

Vertexes and solutions:



Linear Programs and Their Geometry

Linear programs (also holds for **max**):

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Canonical form

$$\min_{x \in \mathbb{R}^n} \langle c, x \rangle$$

$$Ax = b$$

$$x_i \geq 0, \quad i \in I \subset \{1, \dots, n\}$$

Vertexes and solutions:

$x' \in \mathbb{R}^n$ is a vertex of a polyhedron P if there exists $c \in \mathbb{R}^n$ such that

$$\max_{x \in P} \langle c, x \rangle$$

is finite and has a unique solution

N-Dimensional Simplex

$$\Delta^n = \{p \in \mathbb{R}_+^n : \sum_{i=1}^n p_i = 1\}$$

Examples: 1) Yes~~✓~~ 2) No

0.3; 0.1; 0.7; 0.0;

N-Dimensional Simplex

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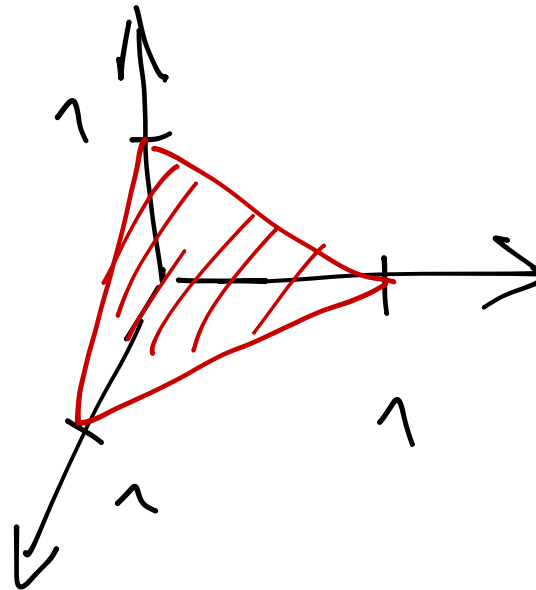
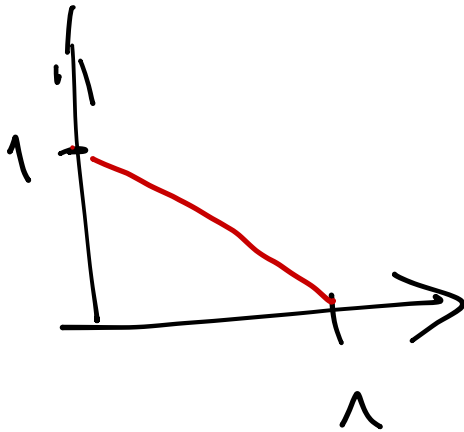
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N-Dimensional Simplex

$$\Delta^n = \{p \in \mathbb{R}_+^n : \sum_{i=1}^n p_i = 1\}$$

Draw your own picture:



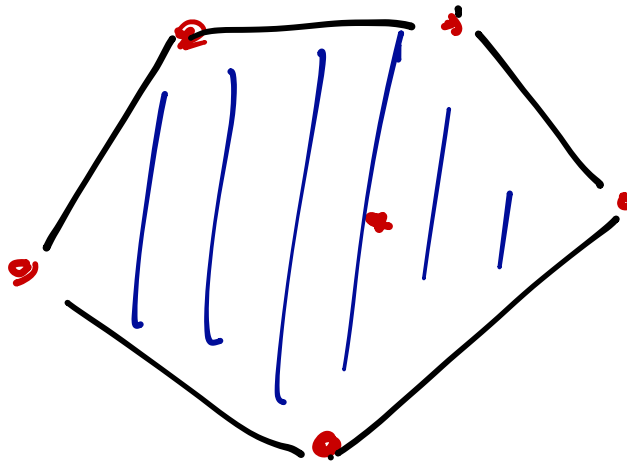
Convex Hull

$\delta^i \in \mathbb{R}^N$, $i = 1, \dots, n$ - arbitrary vectors

Convex hull: **Always a polytope!**

$$\text{conv}\{\delta^i : i = 1, \dots, n\} := \left\{ s : s = \sum_{i=1}^n p_i \delta^i, p \in \Delta^n \right\}$$

Picture:



Theorem 1:

Polyhedron P is a polytope if and only if it is representable
as a convex hull of its vertexes.

Theorem 2:

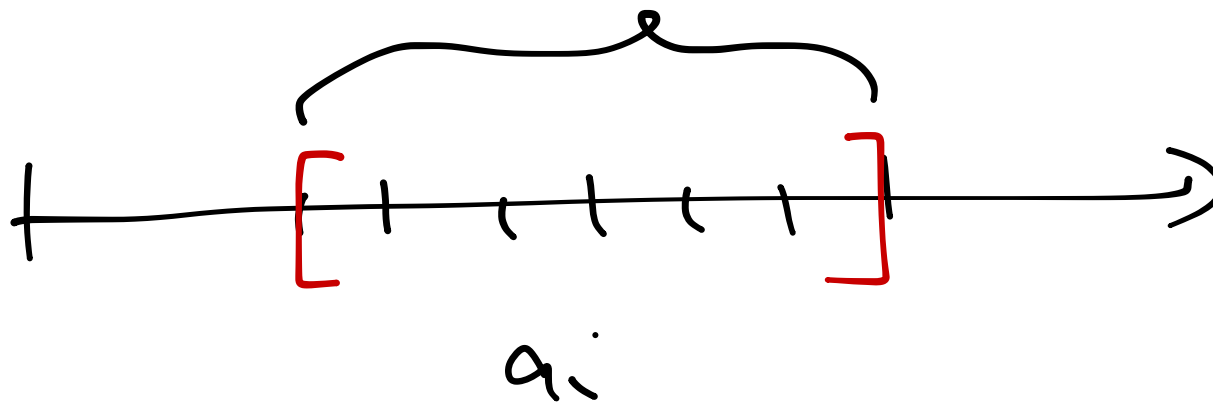
Let $\delta \in \mathbb{R}^N$ be a vertex of $\text{conv}\{\delta^i : i = 1, \dots, n\}$

Then there exists $i \in \{1, \dots, n\}$ such that $\delta = \delta^i$.

Convex Hull

Lemma 1:

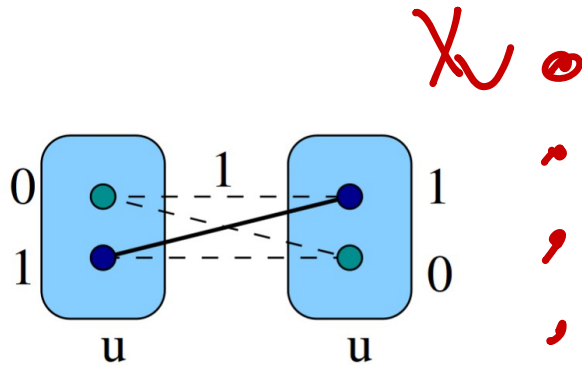
$$\min_{i=1,\dots,n} \{a_i\} = \min_{p \in \Delta^n} \sum_{i=1}^n p_i a_i = \min_{\mu \in \text{conv}\{a_i, i=1,\dots,n\}} \mu$$



Marginal Polytope: Labeling $\rightarrow$ Vector

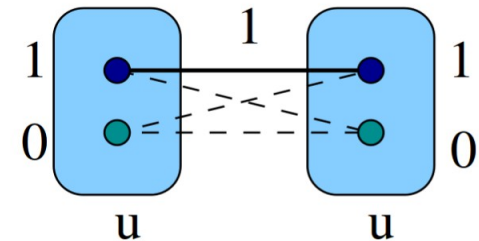
Mapping: labeling $\rightarrow$ binary vector

$$\delta: X_V \rightarrow \{0, 1\}^I$$



$$\delta(x^1) = \underbrace{01}_u \underbrace{0100}_{uv} \underbrace{10}_v$$

$Ax \leq b$

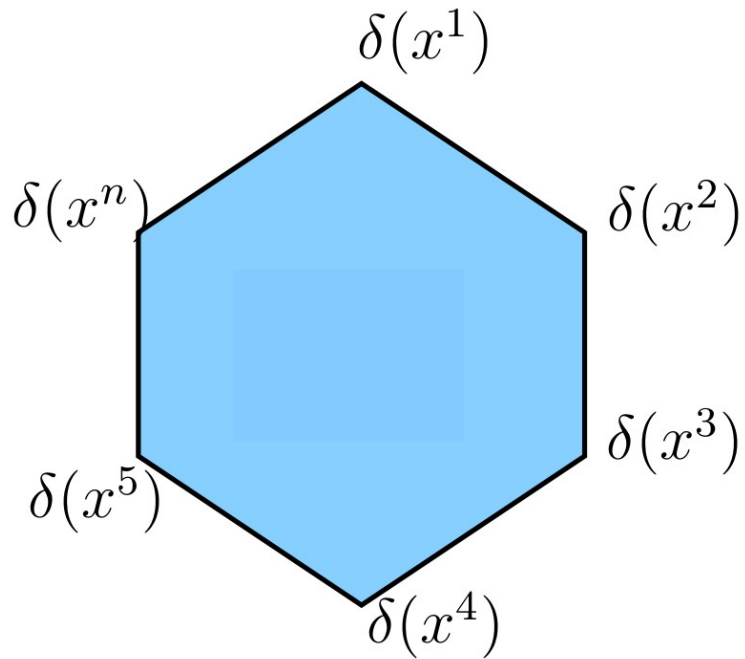


$$\delta(x^2) = \underbrace{10}_u \underbrace{1000}_{uv} \underbrace{10}_v$$

$$\min_{x \in X_V} \sum_{v \in V} \theta_v(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}(x_u, x_v) \equiv \min_{x \in X_V} \langle \theta, \delta(x) \rangle$$

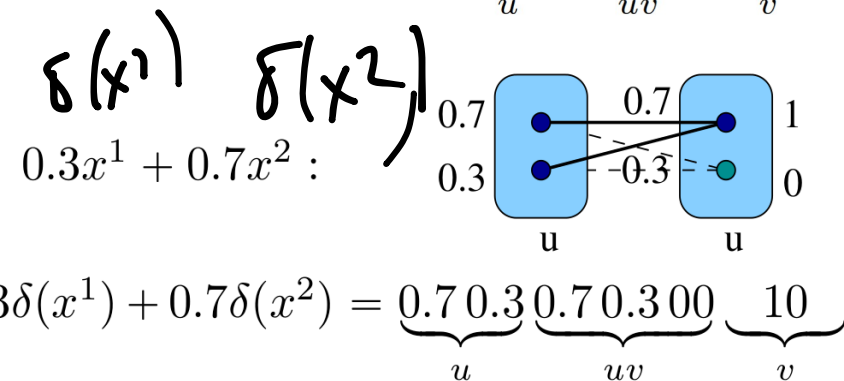
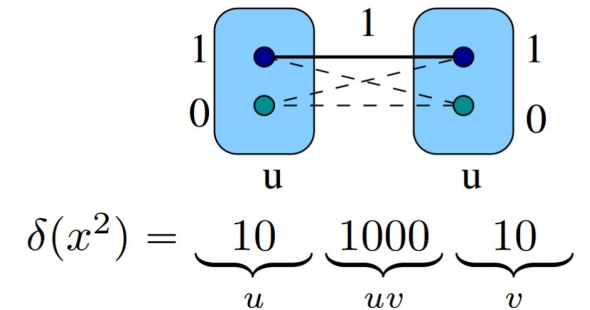
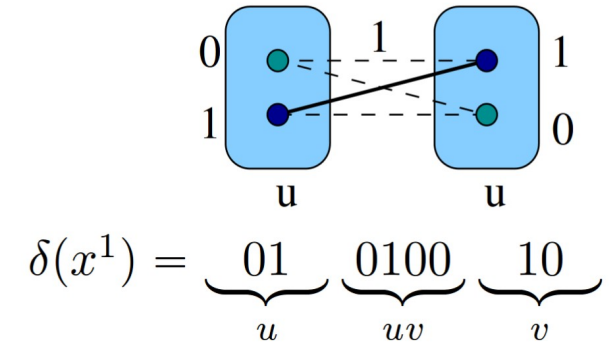
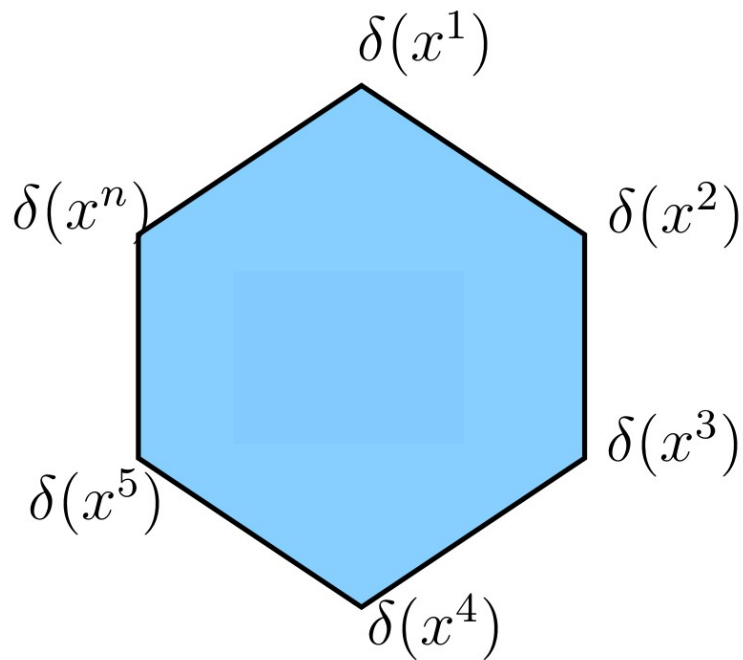
Marginal Polytope

Marginal polytope: $M := \text{conv}\{\delta(x) \in \mathbb{R}^I : x \in X_V\}$



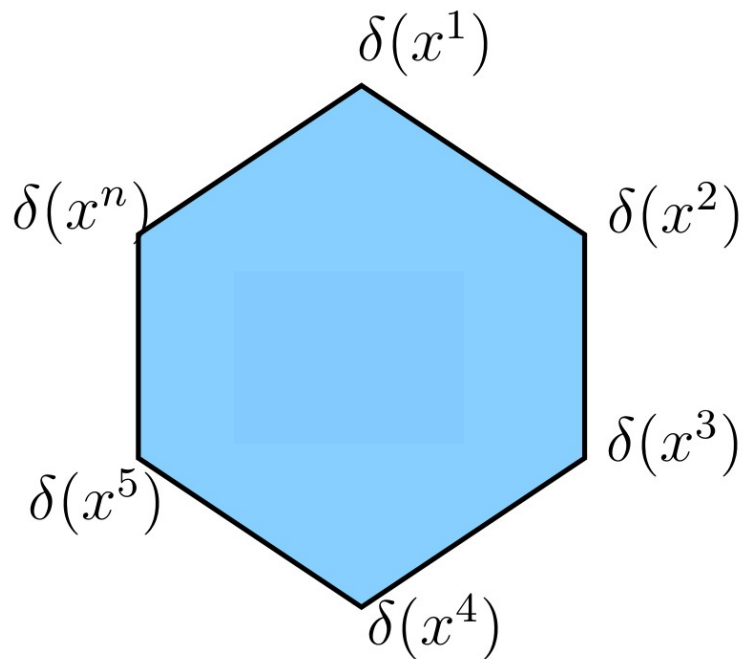
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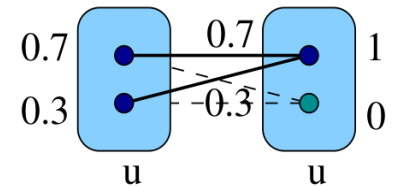


Corollary 1:

$\mu_v \in \Delta^{|X_v|}$ for any $v \in V$

$\mu_{uv} \in \Delta^{|X_{uv}|}$ for any $uv \in \mathcal{E}$

$0.3x^1 + 0.7x^2 :$



$$0.3\delta(x^1) + 0.7\delta(x^2) = \underbrace{0.7 \ 0.3}_u \underbrace{0.7 \ 0.3 \ 0 \ 0}_{uv} \underbrace{10}_v$$

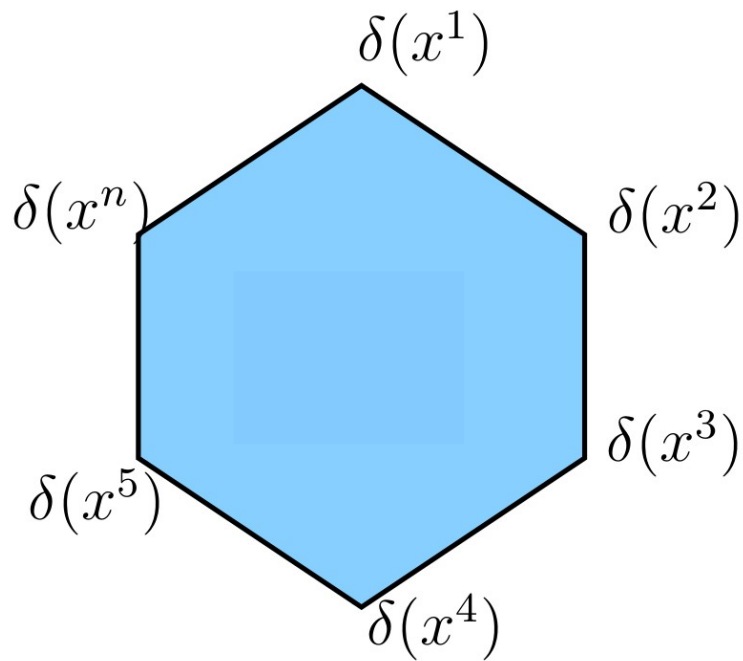
Marginal Polytope

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Proposition 2:

$$\min_{x \in X_V} \langle \theta, \delta(x) \rangle = \min_{\mu \in M} \langle \theta, \mu \rangle$$

◻



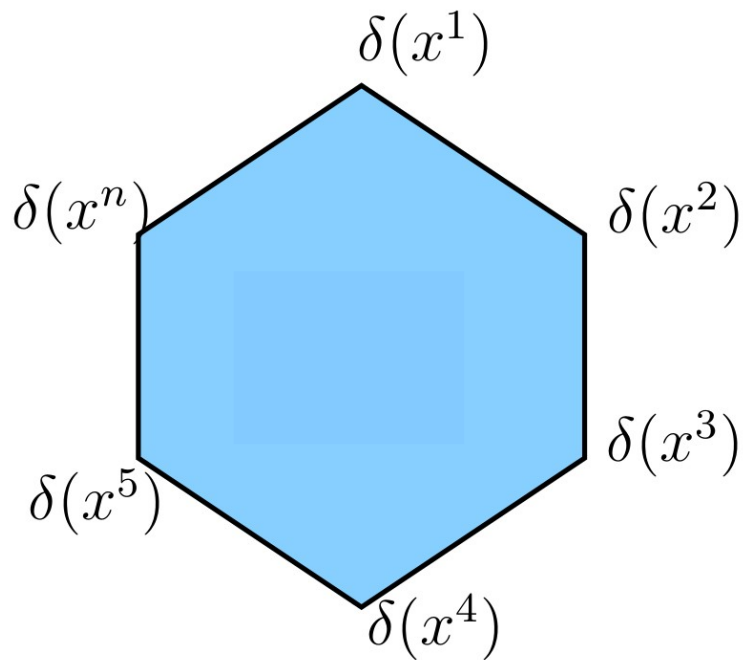
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NP-hard problem = linear problem?



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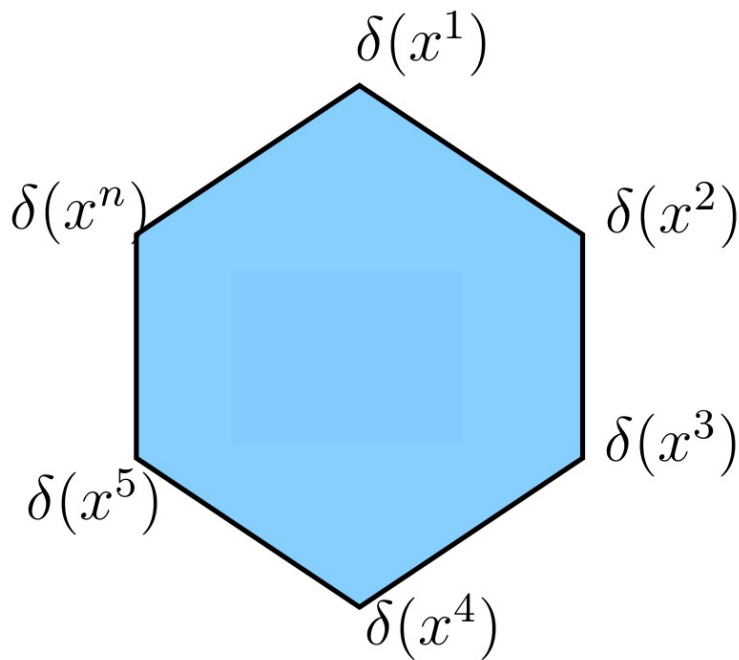
NP-hard problem = linear problem?

$$\sum_{x \in X_V} p_x \delta(x) = \mu$$

$$\sum_{x \in X_V} p_x = 1$$

$$p_x \geq 0, \quad x \in X_V$$

Yes, with an exponential number of constraints!



Marginal Polytope

Marginal polytope: $M := \text{conv}\{\delta(x) \in \mathbb{R}^I : x \in X_V\}$

Proposition 3:

For any $x \in X_V$

the vector $\delta(x)$

is a vertex of M .

